

Iron Deficiency Anemia

Iron deficiency anemia is the most common nutritional deficiency worldwide. It develops when iron stores fall below the level required for normal erythropoiesis, most often through chronic blood loss, inadequate dietary intake, or impaired absorption in the duodenum. Presenting symptoms include fatigue, pallor, exertional dyspnoea and, in severe cases, pica. Diagnosis rests on a low serum ferritin with reduced transferrin saturation. Treatment is oral ferrous sulfate together with investigation and correction of the underlying source of loss.

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Diabetes Mellitus Type 2

Type 2 diabetes mellitus is characterised by insulin resistance with relative insulin deficiency. Type 2 diabetes mellitus is characterised by insulin resistance with relative insulin deficiency. Diagnosis is established by a fasting plasma glucose of 126 mg per deciliter or higher on two occasions, a two-hour value of 200 mg per deciliter on an oral glucose tolerance test, or a glycated hemoglobin of 6.5 percent or higher. Initial management combines dietary modification and physical activity with metformin as first-line pharmacotherapy. Long-term complications include retinopathy, nephropathy, peripheral neuropathy and accelerated atherosclerosis.

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Appendicitis

Appendicitis is inflammation of the vermiform appendix, usually following luminal obstruction by a fecalith or lymphoid hyperplasia. The classic presentation is periumbilical pain migrating to the right lower quadrant, accompanied by anorexia, nausea and low-grade fever. Rebound tenderness at McBurney point is characteristic on examination. Untreated, the appendix may perforate within twenty-four to seventy-two hours, producing localised abscess or generalised peritonitis. Definitive treatment is appendectomy, performed laparoscopically where possible.

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WOUND HEALING

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Wound healing proceeds through four overlapping phases. Hemostasis begins within seconds as platelets aggregate and a fibrin clot forms. The inflammatory phase follows, with neutrophils and then macrophages clearing debris and bacteria over the first three days. Proliferation spans roughly day three to day twenty-one, during which fibroblasts deposit collagen, new capillaries form and epithelial cells migrate across the defect. Remodeling continues for up to a year as type III collagen is replaced by stronger type I collagen.

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HYPERTENSION

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Immunization

Vaccines produce immunity by presenting the immune system with antigen in a form that does not cause disease. B lymphocytes generate antibody while memory B and T cells persist, allowing a faster and larger secondary response on later exposure. Live attenuated vaccines generally produce more durable immunity than inactivated preparations but are contraindicated in immunocompromised patients. Herd immunity protects those who cannot be vaccinated when a sufficient proportion of the population is immune.

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VITAMINS

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Vitamins are organic compounds required in small amounts that the body cannot synthesise in sufficient quantity. The fat-soluble vitamins are A, D, E and K; they are stored in hepatic and adipose tissue and can accumulate to toxic concentrations. Water-soluble vitamins comprise the B-complex group and vitamin C, which are excreted in urine and require regular replacement. Deficiency states include night blindness with vitamin A, rickets and osteomalacia with vitamin D, and scurvy with prolonged vitamin C deficiency.

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